

```
import pandas as pd
import numpy as np

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix

data = pd.read_csv("bill_authentication.csv")
print(data.head())
```

	Variance	Skewness	Curtosis	Entropy	Class
0	3.62160	8.6661	-2.8073	-0.44699	0
1	4.54590	8.1674	-2.4586	-1.46210	0
2	3.86600	-2.6383	1.9242	0.10645	0
3	3.45660	9.5228	-4.0112	-3.59440	0
4	0.32924	-4.4552	4.5718	-0.98880	0

```
x = data.drop('Class' , axis = 1)
print(x)
```

	Variance	Skewness	Curtosis	Entropy
0	3.62160	8.66610	-2.8073	-0.44699
1	4.54590	8.16740	-2.4586	-1.46210
2	3.86600	-2.63830	1.9242	0.10645
3	3.45660	9.52280	-4.0112	-3.59440
4	0.32924	-4.45520	4.5718	-0.98880
...	...	...	...	...
1367	0.40614	1.34920	-1.4501	-0.55949
1368	-1.38870	-4.87730	6.4774	0.34179
1369	-3.75030	-13.45860	17.5932	-2.77710
1370	-3.56370	-8.38270	12.3930	-1.28230
1371	-2.54190	-0.65804	2.6842	1.19520

[1372 rows x 4 columns]

```
x = data.drop('Class' , axis = 1)
print(x)
```

	Variance	Skewness	Curtosis	Entropy
0	3.62160	8.66610	-2.8073	-0.44699
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```
1369 -3.75030 -13.45860 17.5932 -2.77710
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1371 -2.54190 -0.65804 2.6842 1.19520
```

```
[1372 rows x 4 columns]
```

```
y = data['Class']
print(y)
```

```
0      0
1      0
2      0
3      0
4      0
```

```
..
1367    1
1368    1
1369    1
1370    1
1371    1
```

```
Name: Class, Length: 1372, dtype: int64
```

```
x_train, x_test, y_train, y_test = train_test_split(x, y,
test_size=0.20)
```

```
model = SVC(kernel='sigmoid')
model.fit(x_train, y_train)
y_pred = model.predict(x_test)
```

```
print("Linear Kernel SVM Accuracy:", accuracy_score(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test,
y_pred))
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n", cm)
```

```
Linear Kernel SVM Accuracy: 0.6945454545454546
```

```
Classification Report:
```

	precision	recall	f1-score	support
0	0.72	0.69	0.71	147
1	0.66	0.70	0.68	128
accuracy			0.69	275
macro avg	0.69	0.69	0.69	275
weighted avg	0.70	0.69	0.69	275

```
Confusion Matrix:
```

```
[[102  45]
 [ 39  89]]
```

```
#test size = 30
```

```
x_train, x_test, y_train, y_test = train_test_split(x, y,
test_size=0.30)
model = SVC(kernel='sigmoid')
model.fit(x_train, y_train)
y_pred = model.predict(x_test)

print("Linear Kernel SVM Accuracy:", accuracy_score(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test,
y_pred))
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n", cm)
```

Linear Kernel SVM Accuracy: 0.6796116504854369

Classification Report:

	precision	recall	f1-score	support
0	0.72	0.71	0.71	233
1	0.63	0.64	0.64	179
accuracy			0.68	412
macro avg	0.67	0.68	0.67	412
weighted avg	0.68	0.68	0.68	412

Confusion Matrix:

```
[[165  68]
 [ 64 115]]
```

```
#test size = 35
```

```
x_train, x_test, y_train, y_test = train_test_split(x, y,
test_size=0.35)
model = SVC(kernel='sigmoid')
model.fit(x_train, y_train)
y_pred = model.predict(x_test)

print("Linear Kernel SVM Accuracy:", accuracy_score(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test,
y_pred))
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n", cm)
```

Linear Kernel SVM Accuracy: 0.656964656964657

Classification Report:

	precision	recall	f1-score	support
0	0.66	0.73	0.69	257
1	0.65	0.58	0.61	224
accuracy			0.66	481

macro avg	0.66	0.65	0.65	481
weighted avg	0.66	0.66	0.65	481

Confusion Matrix:

```
[[187  70]
 [ 95 129]]
```

#test size = 40

```
x_train, x_test, y_train, y_test = train_test_split(x, y,
test_size=0.40)
```

```
model = SVC(kernel='sigmoid')
```

```
model.fit(x_train, y_train)
```

```
y_pred = model.predict(x_test)
```

```
print("Linear Kernel SVM Accuracy:", accuracy_score(y_test, y_pred))
```

```
print("Classification Report:\n", classification_report(y_test,
y_pred))
```

```
cm = confusion_matrix(y_test, y_pred)
```

```
print("Confusion Matrix:\n", cm)
```

Linear Kernel SVM Accuracy: 0.7213114754098361

Classification Report:

	precision	recall	f1-score	support
0	0.76	0.75	0.76	315
1	0.67	0.68	0.68	234

accuracy			0.72	549
macro avg	0.72	0.72	0.72	549
weighted avg	0.72	0.72	0.72	549

Confusion Matrix:

```
[[237  78]
 [ 75 159]]
```